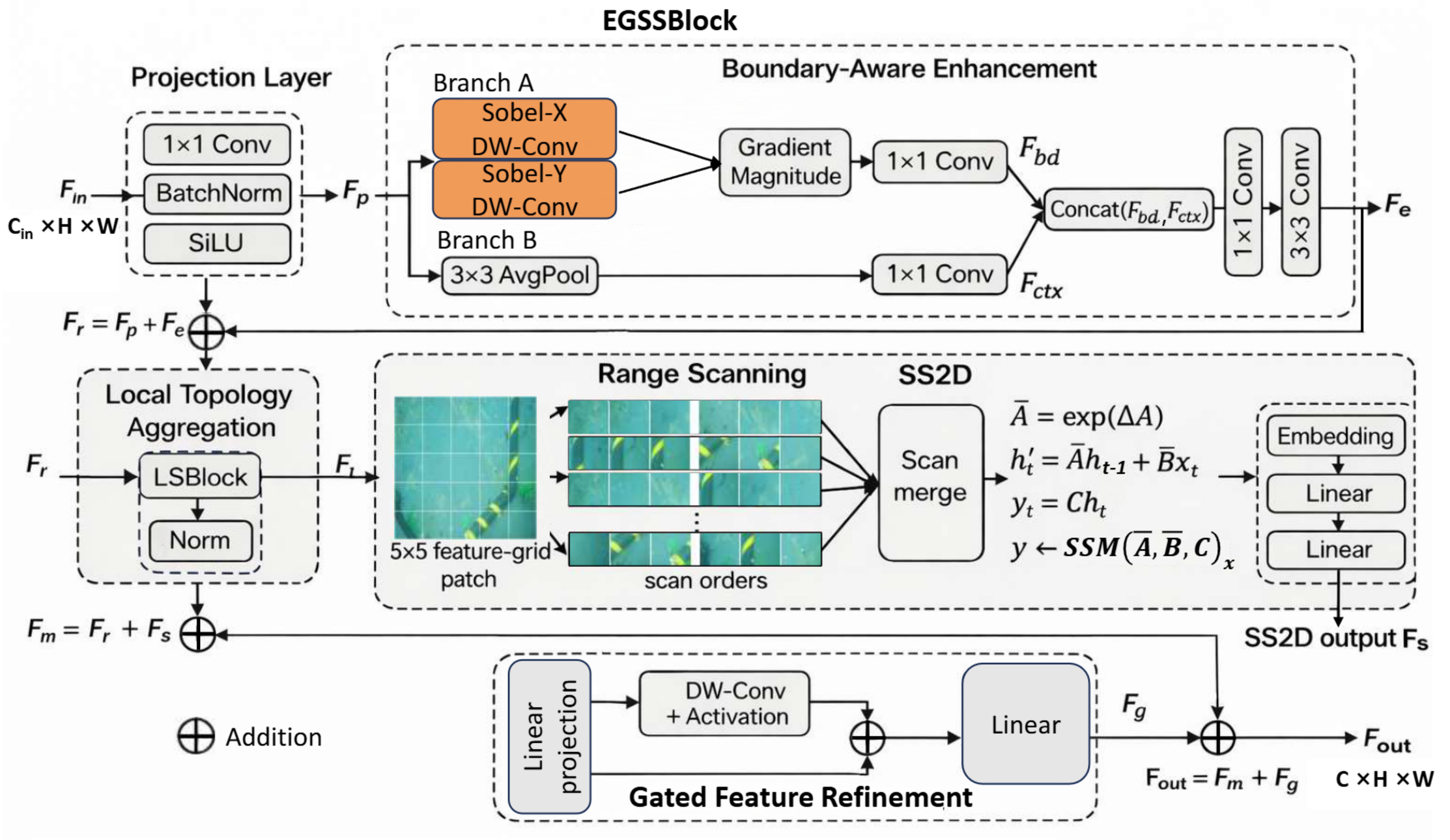
EGSSBlock
Projection Layer
1×1 Conv
BatchNorm
SiLU
F_{in}
$C_{in} \times H \times W$
F_p
Boundary-Aware Enhancement
Branch A
Sobel-X DW-Conv
Sobel-Y DW-Conv
Gradient Magnitude
1×1 Conv
F_{bd}
Branch B
3×3 AvgPool
1×1 Conv
F_{ctx}
Concat(F_{bd}, F_{ctx})
1×1 Conv
3×3 Conv
F_e
$F_r = F_p + F_e$
Local Topology Aggregation
F_r
LSBlock
Norm
F_l
Range Scanning
SS2D
5×5 feature-grid patch
scan orders
Scan merge
$\bar{A} = \exp(\Delta A)$
$h'_t = \bar{A}h_{t-1} + \bar{B}x_t$
$y_t = Ch_t$
$y \leftarrow SSM(\bar{A}, \bar{B}, C)_x$
Embedding
Linear
Linear
SS2D output F_s
$F_m = F_r + F_s$
Addition
Linear projection
DW-Conv + Activation
Linear
Gated Feature Refinement
F_g
F_{out}
$F_{out} = F_m + F_g$
$C \times H \times W$